FIT5196-S1-2026 assessment 1 (35%)

This is a group assessment and worths 35% of your total mark.

Due date: 23:55, Thursday, Week 7 (23 April 2026)

Exploratory Data Analysis (EDA) plays a critical role in the broader data analysis lifecycle. It acts as the bridge between raw data and actionable insights, allowing analysts to assess data quality, detect errors, and identify potential transformations needed to prepare the data for modelling. In the context of data wrangling, EDA guides the cleaning and reshaping process by revealing inconsistencies, duplicates, and missing values that must be addressed. Moreover, the insights generated through EDA inform data-driven decision-making by highlighting key metrics and revealing patterns that support evidence-based strategies. Effective EDA ensures that subsequent analysis is grounded in a solid understanding of the dataset and increases the reliability and interpretability of any models or decisions derived from the data.

Assessment Objectives

In this group assessment, students will undertake an Exploratory Data Analysis (EDA) project as part of the Data Wrangling unit. The primary objective of EDA is to develop a comprehensive understanding of a dataset before engaging in further modelling or machine learning processes. Through this process, students will identify trends, patterns, anomalies, and relationships within the data that may inform or inspire meaningful machine learning research questions.

You are expected to gain knowledge, skills and experience on

- **Data Familiarisation**: Gain a deep and structured understanding of the dataset's structure, content, and quality.
- **Basic Data Quality Assessment**: Identify missing values, duplicates, invalid values or inconsistencies that may affect analysis, but no need to fix them.
- **Visual Exploration**: Use visualisation techniques to uncover hidden structures, correlations, and distributions in the data.
- **Research Insight Generation**: Translate EDA findings into potential research questions that could be explored further using machine learning approaches.

To achieve the above objectives, the following step-by-step guide is recommended:

Step 0: Grouping and Special Considerations
Please form a group of 4 - 5 students well ahead before the group selection deadline - March 30, 2026.

Note that 2-day short extensions (aka regular extensions) are not applicable to group assessments. If you cannot submit on time, please check whether you are qualified for a special consideration (Link). Note that special consideration applications are ***not*** managed by the Faculty teaching team and you should contact the University special considerations team directly.

Step 1: Load, parse and merge data files

Each group has **unique** input data files (link) to work with. Please note that using wrong input data files may lead to wrong findings from EDA and potentially result in **SIGNIFICANT** mark deduction for the assessment. Please double check that you have the correct input data files matching with your group number.

Input data files:

- Group_<group_number>.xml
- Group_<group_number>.json

These two files contain photo (meta) data collected from Flickr. Each photo is encapsulated in a record that contains 18 attributes. Please check with the sample input files (sample_input) for all the available attributes.

Your tasks:

- Use appropriate Python codes to load and extract data from both files;
- Convert loaded data into suitable objects for ease of manipulation;
- Inspect the structure and schema of data from both files to understand their components and relationship;
- Merge data into a single dataset using appropriate methods taught in the unit;
- Ensure the merged data is well structured, free from redundant duplicate records, and formatted correctly.
- Any non-Latin and non-digit content are required to be appropriately handled with good reasons and justifications conforming to data wrangling practices. Requirements:
    - All text data in these five (5) attributes: *Title*, *City*, *Country*, *Tags*, *Description*, must be transformed into lowercase ('NaN' excluded);
    - No XML or HTML tags, no emojis, only valid UTF-8 characters are allowed (you must only use Regular Expression for this task);
    - Non-Latin characters need to be **completely removed** from all the columns (you must only use Regular Expression for this task). As an example, Japanese, Chinese, and Russian Cyrillic alphabets should be removed, but you should still retain European languages using latin characters.
    - All null values need to be represented by 'NaN'.
- Output the valid merged data to a CSV file named Group<group_number>_dataset.csv (see sample_output_file). Your generated CSV file should match the sample output file.

Note: You are required to identify appropriate **Regular Expressions** to process the text data as required above. You need to have the exactly **same attributes** in your CSV file as that in the sample output file. Otherwise, marks may be deducted.

Note: Your Python solution will be tested & assessed by our markers using a **test dataset** (hidden from students) to check for its correctness. In practice, that means we require your code to work with **any future or unknown dataset** that has the same structure.

<u>Warning:</u> A solution with bugs will often lead to mark deduction. Please debug your code properly before submission. Markers will not fix your code or handle any data/coding errors during the marking / assessment phase(s).

Step 2: Apply EDA on the merged dataset with visualizations

The primary goal of Exploratory Data Analysis (EDA) is to **gain an in-depth understanding of the dataset** to inform subsequent decisions, such as data preprocessing, model design, or hypothesis generation.

EDA helps you to:

- Identify **structure and patterns** within the data
- Detect **errors, outliers, or anomalies**
- Assess the **quality and completeness** of the data
- Generate **insights and hypotheses** to guide further analysis, such as machine learning tasks

Your tasks:

- Understand the overall structure of the merged dataset (Hint: Basic sanity checks such as dimensions, record numbers, sizes, and all other relevant techniques we taught during the applied classes.)
- Apply univariate analysis
- Apply bivariate analysis
- Apply multivariate analysis
- Visualise the findings

By the end of this EDA process, you should be able to

- Describe the key characteristics of the dataset
- Explain the relationships and trends observed
- Identify data issues that must be addressed before modelling
- Create appropriate visualizations to present your understanding of the data and valuable findings, and capable of justifying the chosen visualizations against other choices.

Step 3: Summarise key insights and research questions

To identify key insights and high quality research questions, you need to review the findings from EDA and translate them into practical and reasonable machine learning problems.

You should consider a number of possible machine learning questions, including

- Supervised learning - classification and regression
- Unsupervised learning - clustering, dimensionality reduction
- Semi-supervised learning
- Others - time series/sequential pattern detection, anomaly detection, etc.

Tips: What types of machine learning problems/questions have you learned in classes? What are the suitable and reasonable types to consider? Are the formulated problems/questions reasonable in practice? Are they useful to the relevant industry?

For this step, at least ten (10) key (high-quality) insights and five (5) high-quality ML questions are required.

A well-formulated machine learning research question should be data driven, which is rooted in actual EDA findings, not speculative or hypothetical (i.e., you need to have solid evidence from your EDA and corresponding visualizations to justify the machine learning research question is reasonable). The ML question should be carefully defined and explained clearly.

For classification questions, what are the input features and output variables? Have you explained them clearly, and do they (input & output) make sense? Make sure your explanations are solid, reasonable, and sound. To avoid drafting low-quality insights/ML questions, please try to think logically. If you want to predict user emotions, is it even sound/reasonable to try to predict user emotion without any sentiment or behavioural data in the dataset? All the provided ML questions should be able to generate value, that is, by studying these questions, one can contribute meaningfully to the research domain or business context.

To evaluate the quality of your ML questions, you can ask yourself the following:

- What variables do I want to predict or understand?
- What predictors do I have available?
- Is there a meaningful relationship supported by the EDA?
- What ML technique is best suited to this problem?
- What would success look like in this task (e.g., high accuracy, good clustering)?

Note: You need to consider all the attributes in EDA, identifying findings and ML questions.

Examples of low-quality (e.g., vague, unclear, or irrelevant) insights and ML questions:

- "The value of that attribute/parameter/variable is small."
- "The trend of that attribute/parameter/variable is increasing/decreasing."
- "Can we use ML to learn something more?"
- "Can we use ML to learn all the attributes/parameters?"

In general, these types of insights/ML questions are either too general, unclear, not related to the business/industry, do not link to any concrete research questions, or do not have sufficient available data to support the training/findings. While it is perfectly fine to describe the value/trend/curiosity in your report/presentation, they themselves are often not sufficient to explain the data and would not be classified as high-quality insights/ML questions.

Warning: Low-quality insights and ML questions will not be counted as high-quality.

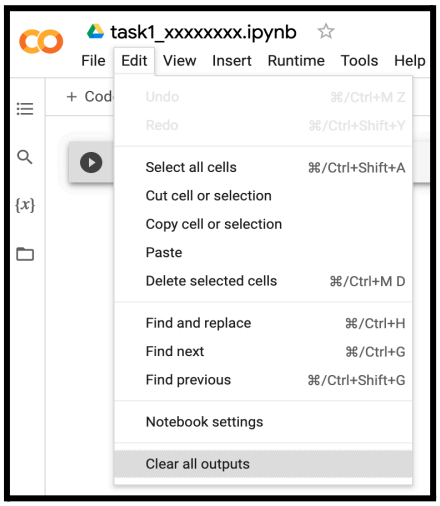
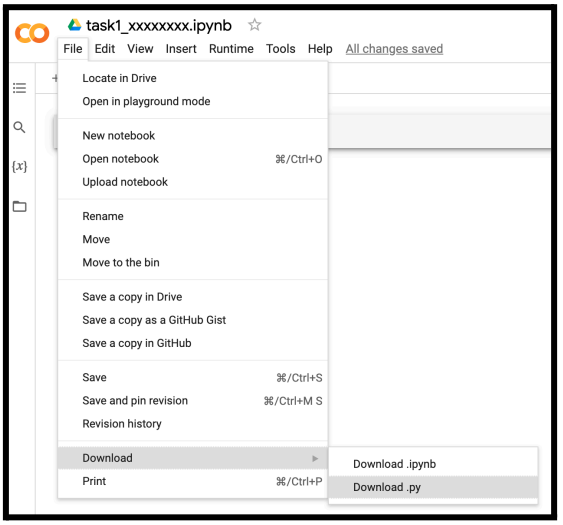
Warning: If markers find students submitting an excessive number of insights or ML questions, markers can choose to only mark the first 10 submitted insights and the first 5 submitted ML questions, and ignore the rest.

Submission Requirements

- **Output and EDA methodology - 30%**
  - A **Group<group_number>_solution.ipynb** file that contains your EDA process and printed outputs (make sure you can use this file to generate the findings/output in your submitted EDA report)
  - A **Group<group_number>_solution.py** file. This file will be used for academic integrity purposes, e.g., plagiarism checks. (make sure you clear your cell output before exporting this .py file).
  - A **Group<group_number>_dataset.csv** file that contains the merged dataset
    Note: Your python codes should be able to produce the same merged dataset as submitted to avoid mark deduction.
- **EDA report - 40%**
  - A **Group<group_number>_EDA.pdf** file that contains your EDA report including your design of the EDA, your findings, insights and ML questions with justifications and visualizations.
- **Video presentation - 25%**
  - A **Group<group_number>_presentation.mp4** file that is a video presentation (6-10 minutes) to effectively communicate your findings from EDA and the ML questions with justification and visualization. All group members need to present a part of the presentation.
- **Generative AI tools declaration and history documentation - 5%**
  - A complete and signed Generative AI Tools Declaration Form (Disclaimer) **Group<group_number>_AI_declaration.pdf**. The template can be found in the Link.
  - A **Group<group_number>_AI_record.pdf** file that contains a complete AI conversation record **in English** (if you have used any AI tools).

Note: Make sure you read the **marking rubric** and understand the criteria before submission.

Note: You can use the **templates** provided for these submission files. Failure to submit a Generative AI Tools Declaration Form or the AI conversation record (if applicable) may lead to mark deduction and potential Academic Integrity cases. The teaching team reserves the rights to put all the submissions through plagiarism detection software(s) that automatically perform similarity checks. Any found plagiarism cases will trigger the Academic Integrity procedures and may result in severe penalties, up to and including exclusion from the university.

Submission Checklist:

- Please zip all the submission files into a single file named **Group<group_number>_A1.zip** except the EDA report file **Group<group_number>_EDA.pdf**. That is, your submission should have two files:
  - **Group<group_number>_A1.zip**
  - **Group<group_number>_EDA.pdf** (if the report pdf file is placed inside the zip file, we may impose an extra 5 marks penalty)
- Make sure all the members in your group click the '**Submit**' button in Moodle
- Please strictly follow the file naming standard.
- Please make sure that your **.ipynb file** contains printed output. However, your **.py file** should not include any output.
- Tips: You can use the options in the menu (for Google Colab users):